DS6021 Project 1: Blue Nile Diamond Data Analysis

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Project 1: Report

Section 1

High Level Analysis Blue Nile, the largest online diamond retailer, claims that "the higher quality a diamond is, the more expensive it will be." Diamond quality is determined by the "4Cs": cut, color, clarity, and carat weight. Our analysis evaluates whether the observed pricing data support this claim.

We began with exploratory data analysis to examine relationships among the variables. The original color and clarity grades were consolidated into broader categories to simplify interpretation. Visualizing carat against price revealed a strong positive relationship, with price increasing at an accelerating rate (polynomial / exponential) as carat weight increased. This nonlinear positive trend supports Blue Nile's claim that larger diamonds relate to higher prices.

Examination of the categorical variables (cut, color, and clarity) using box plots and scatter plots indicated that median prices generally increase with higher quality grades. Diamonds classified as Astor Ideal in cut, Colorless in color, and Flawless in clarity exhibited the highest price distributions. Overall, these patterns align with Blue Nile's stated quality hierarchy, in which superior grades across the 4Cs correspond to values.

To formally assess the relationship between price and carat, we fit a simple linear regression (SLR) model with price as the response and carat as the predictor. The classical SLR assumptions include linearity of the conditional mean (mean error is zero), constant variance, independent, and normally distributed errors. Diagnostic plots from the untransformed model revealed curvature and increasing residual spread, indicating violations of linearity and constant variance. Independence appeared reasonable based on the ACF plot.

A Box-Cox procedure suggested a power transformation of the response with $\lambda = 0.3$. Defining $\hat{Price}^{.3}$ substantially stabilized the variance. Residual curvature remained, motivating a square-root transformation of the predictor ($x^* = \text{carat}^{1/2}$). After applying both transformations, diagnostic plots indicated no systematic curvature and approximately constant variance, suggesting that regression assumptions were adequately satisfied.

The final model is:

$\hat{Price}^{.3} = -2.65220 + 15.91060 * Carat^{1/2}$

The slope is positive and highly statistically significant ($p < 2 * 10^{-16}$), providing strong evidence of a relationship between carat weight and price. Although the power transformations limit direct coefficient interpretation, the results indicate that price increases non linearly with carat weight, with larger diamonds resulting in higher prices. Overall, the findings support Blue Nile's claim that higher quality diamonds, particularly those with greater carat weight, are more expensive, and that carat is a major driver of diamond pricing.

Section 2

Blue Nile Diamond Dataset Overview:

```
##      carat           clarity        color              cut
##  Min.   :0.2300   SI1    :243   F      :223   Good       : 73
##  1st Qu.:0.4000   VS1    :233   D      :207   Very Good  :382
##  Median :0.5200   VS2    :214   G      :198   Ideal      :739
##  Mean   :0.8134   SI2    :165   E      :181   Astor Ideal: 20
##  3rd Qu.:1.0000   VVS2   :158   I      :167
##  Max.   :7.0900   VVS1   :149   H      :148
##                   (Other): 52   (Other): 90
##      price                 color_category                    clarity_category
##  Min.   :   322.0   Faint         :  0    Included                   :  0
##  1st Qu.:   723.5   Near Colorless:603    Slightly Included          :408
##  Median :  1463.5   Colorless     :611    Very Slightly Included     :447
##  Mean   :  7056.7                         Very Very Slightly Included:307
##  3rd Qu.:  4640.8                         Internally Flawless        : 49
##  Max.   :355403.0                         Flawless                   :  3
##
```

Cut measures how well-proportioned a diamond's dimensions are, including its balance and brilliance.

Diamond color refers to how colorless a diamond is.

Clarity assesses small imperfections within a diamond.

Carat measures a diamond's weight, not its size.

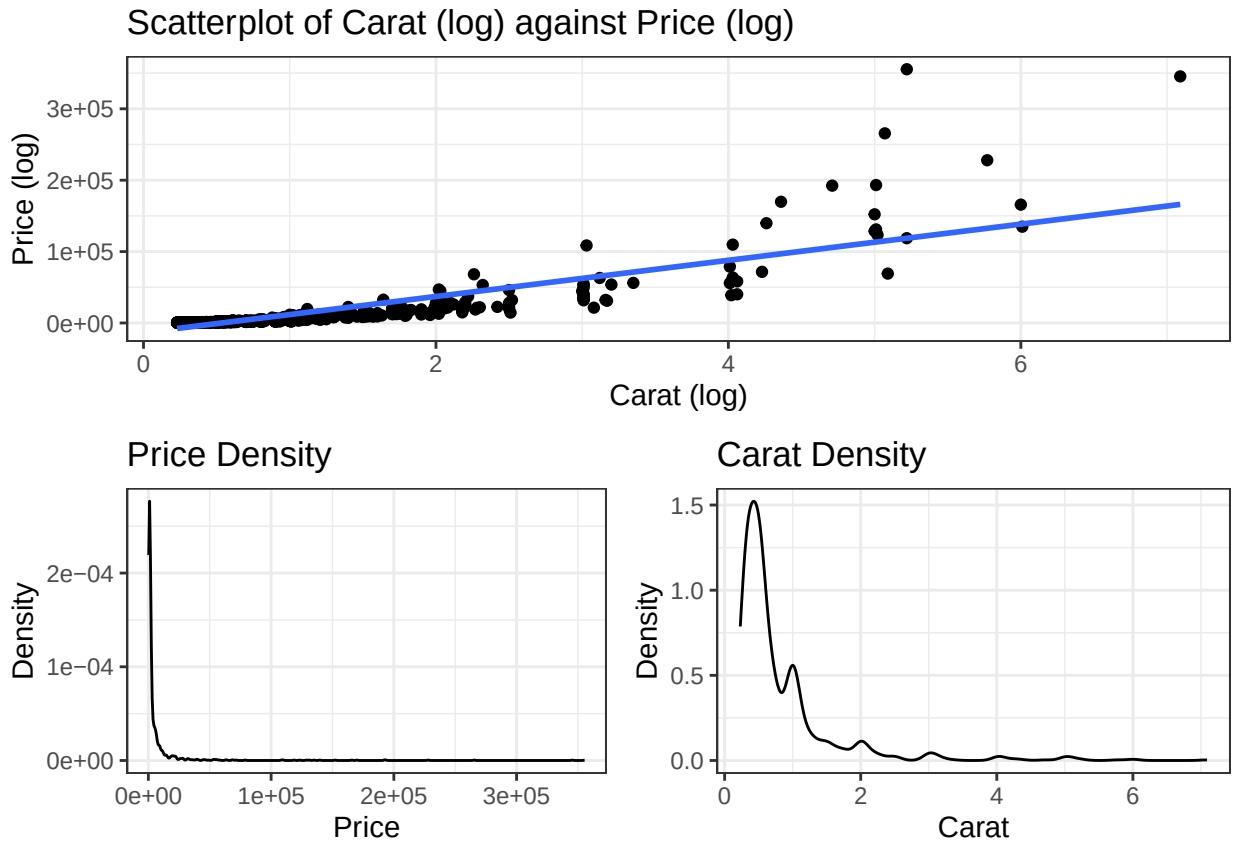
Carat values range from 0.23 to 7, with a median value of 0.52 and mean value of 0.81.

Price values range from 322 to 355403, with a median value of 1463.5 and mean value of 7056.7.

The range of the numerical variables (carat and price) suggest the presence of outlier values that may skew values of central tendency.

Two categorical variables (color_category and clarity_category) were created grouping color and clarity values.

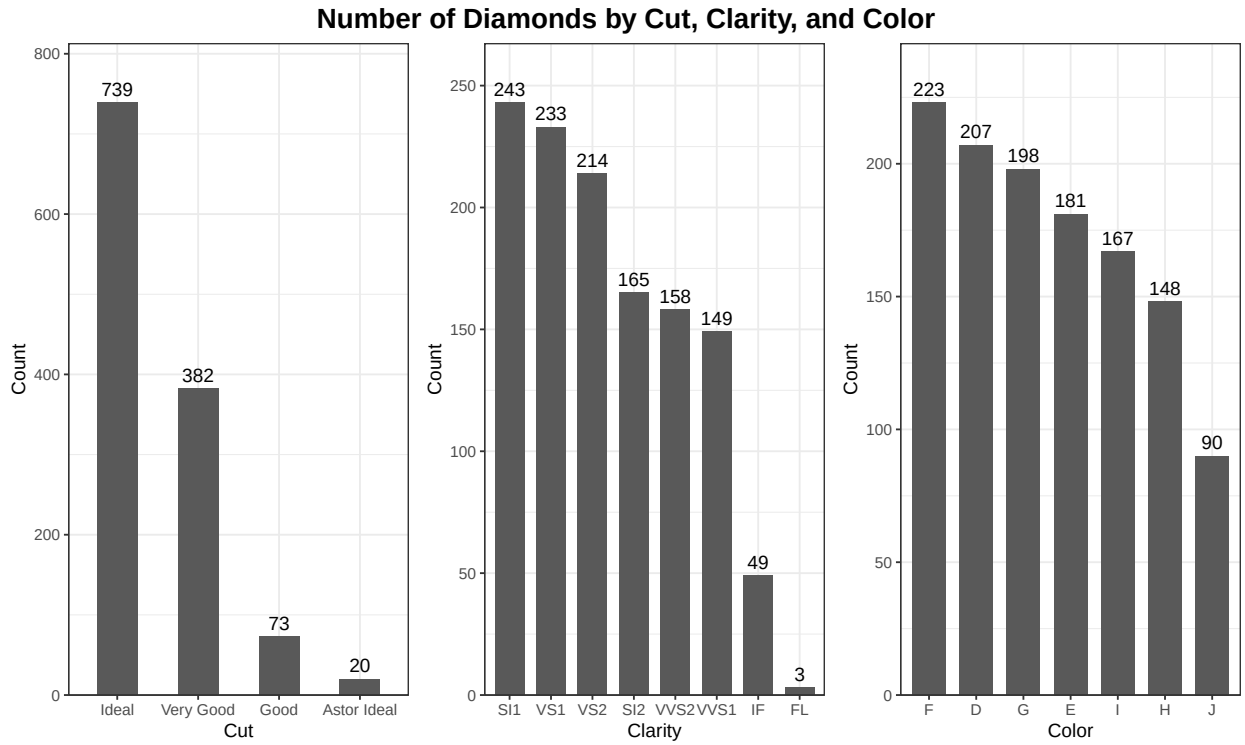
Visualizing the Distribution of Numerical Variables



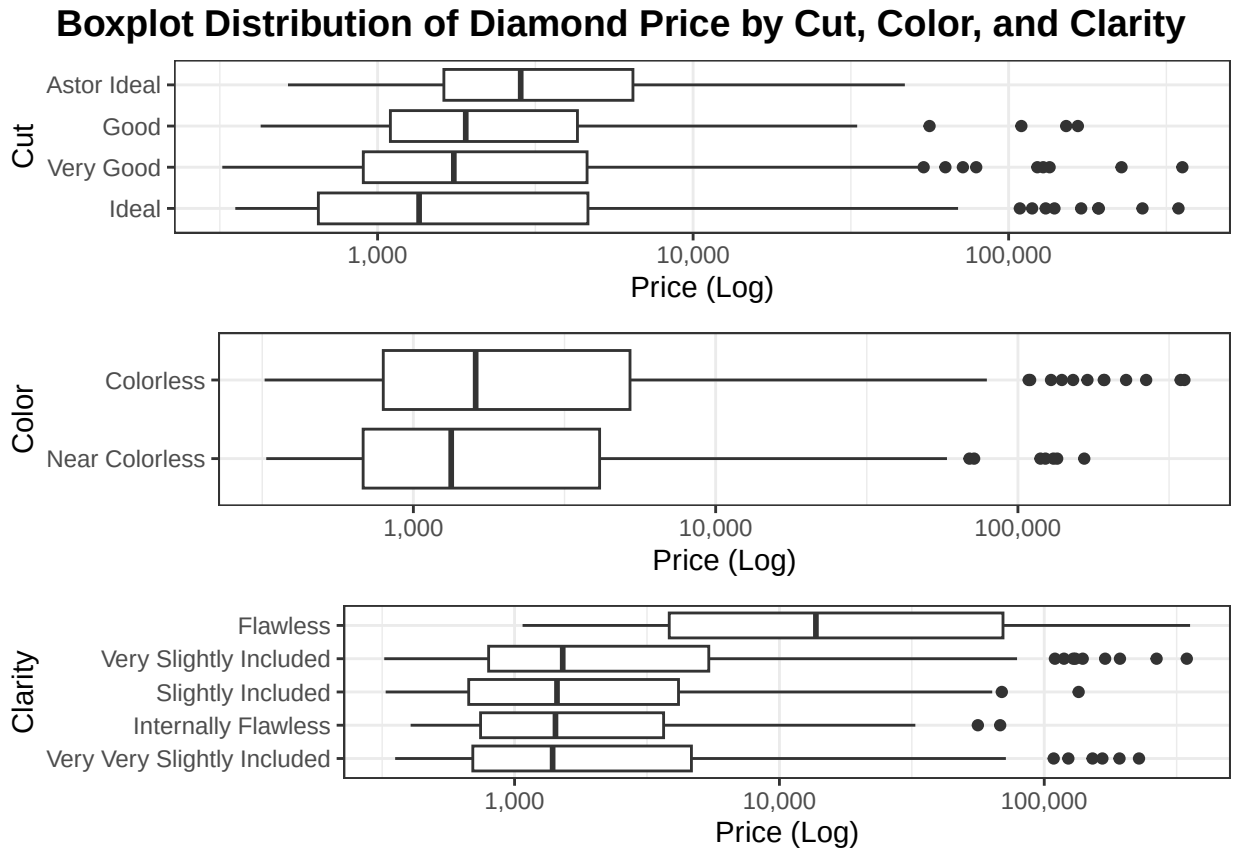
The scatter plot presented here illustrates a strong positive correlation between the carat weight of a diamond and its price.

Density plots for both price and carat demonstrate a right skew indicating that the majority of the diamonds are lower priced and with lower carat weights. The long density tails extending to the right of plot are indicative of outlier diamonds that are rare and expensive.

Visualizing the Distribution of Categorical Variables



Visualizing the Distrubution of Diamond Price by Cut, Color, and Clarity

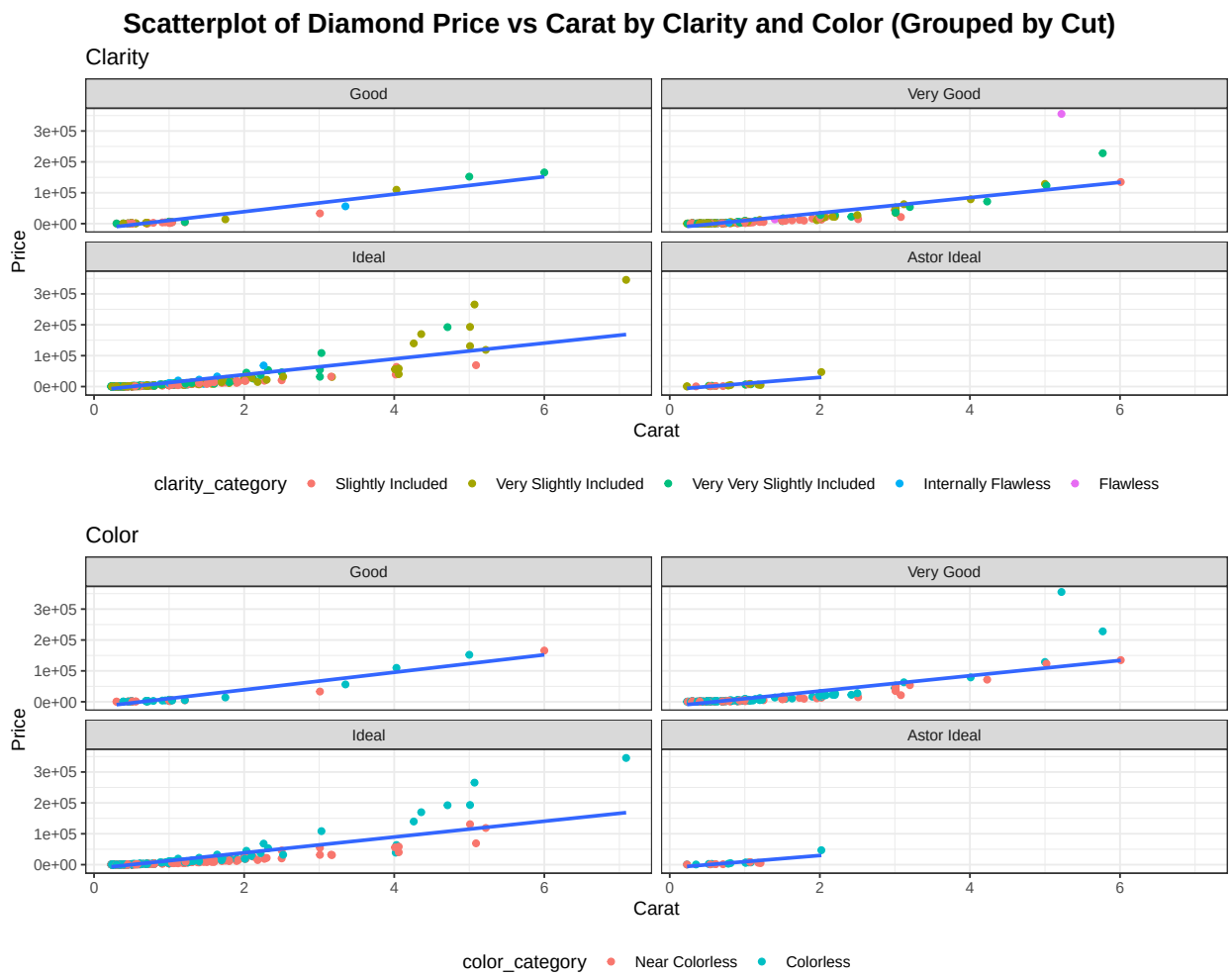


Given the strong right skew of price points in the diamond data set, a log transformation of prices on the x-axis was needed to be performed to better visualize the boxplot distributions of cut, color, and clarity on diamond price.

Regarding diamond cut, Blue Nile claims that diamond cut is the most important of the four Cs (color, carat, clarity, cut), however when looking at the box plot distribution of cut, we can see that diamonds with a good or very good cut classification have a higher median price than "higher quality" ideal cut diamonds.

When looking at diamond color, according to Blue Nile, "part of diamond valuation is determined by the absence of color, with more colorless diamonds receiving higher grades along the diamond color and clarity scale." This suggests that colorless diamonds will be more valuable than near colorless diamonds. From the box plot distribution for color, we can see the data supports this claim as colorless diamonds have a higher median price point than near colorless diamonds.

Lastly, the distribution of prices among clarity grades shows that diamonds considered flawless are the most expensive, while distributions among other clarity classifications are similar. Outside of flawless diamonds, very slightly included and slightly included diamonds appear to have median prices that are slightly more expensive than internally flawless and very very slightly included diamonds. This observation supports Blue Nile's claim that Slightly Included and Very Slightly Included diamonds are good value diamonds.



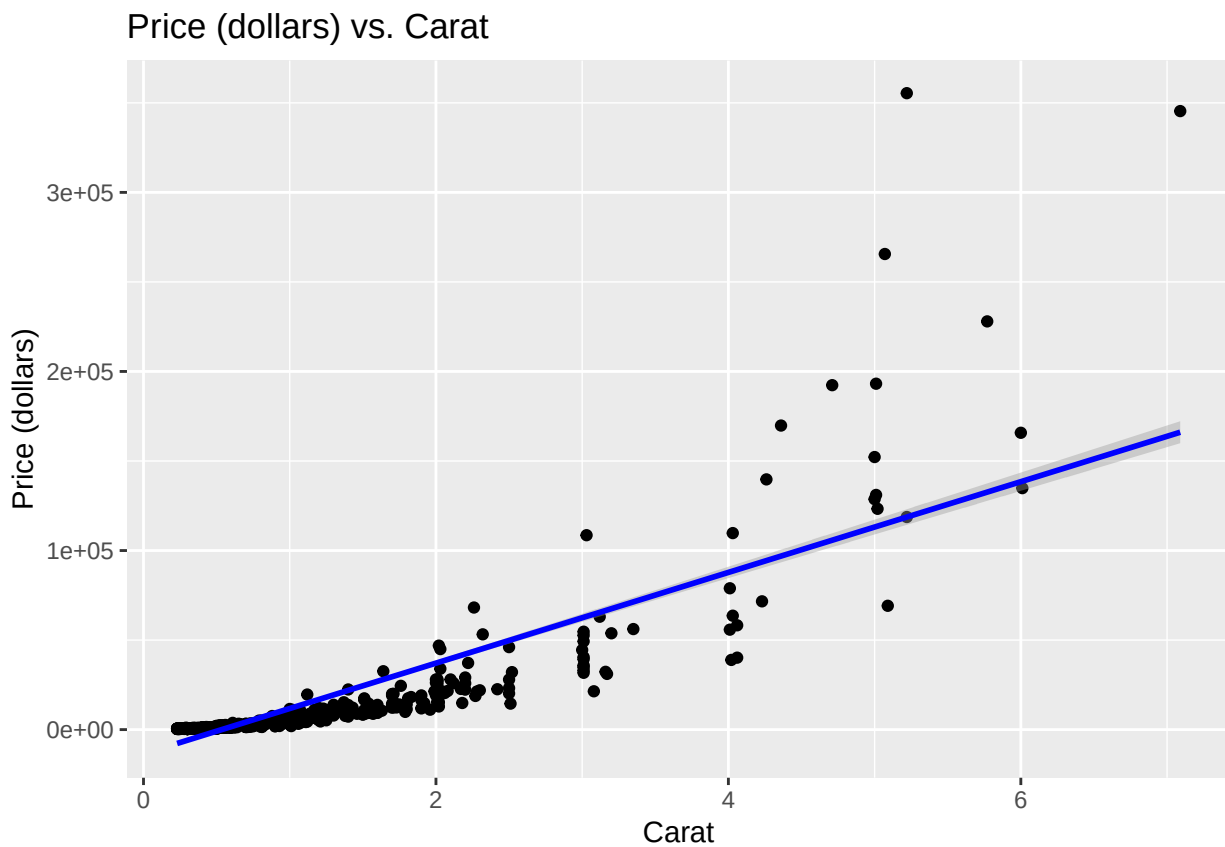
The scatterplot distributions visualized provided added insight in the high median price values of good and very good diamond cuts in the data set. When looking at the clarity scatter plot, we can see very good cuts of diamonds having large carat values with one even having a flawless clarity rating, driving its high price point. Similarly, when looking at the color scatter plot, we can see the very good cuts of diamonds with higher carat weights having colorless classifications, also driving its value higher.

Section 3

Intial Regression Assumption

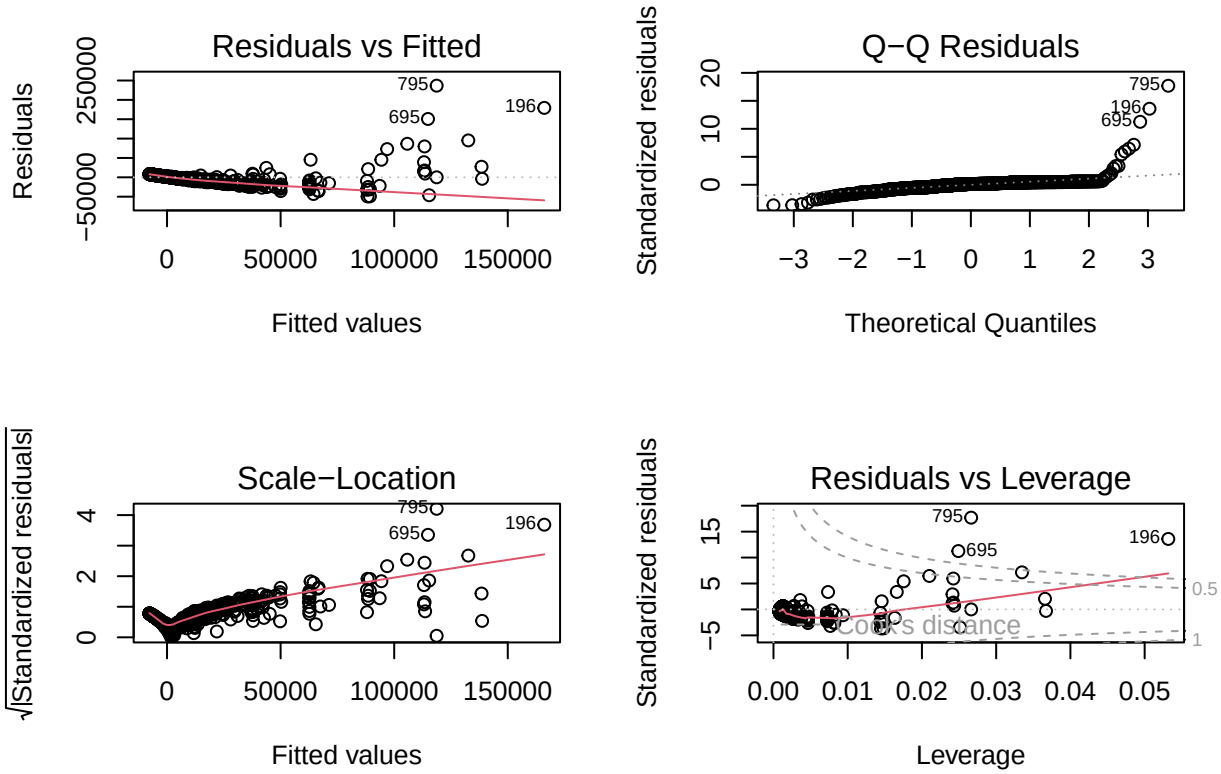
To verify the initial regression assumptions are met, we will examine the scatter plot and residual plot of Price vs. Carat.

```
## `geom_smooth()` using formula = 'y ~ x'
```



In this case, the predictor variable is Carat and the response is Price (dollars). The assumption that mean error is 0 (i.e. linear data) seems not to hold true as the data does not look very linear. For example, the observations of price between 0-.5 and 5-7 of carat is all above the regression line, while the rest of the observations of price are bellow regression line. This means the SLR estimations of price will systemically under predict or over predict price in those respective regions.

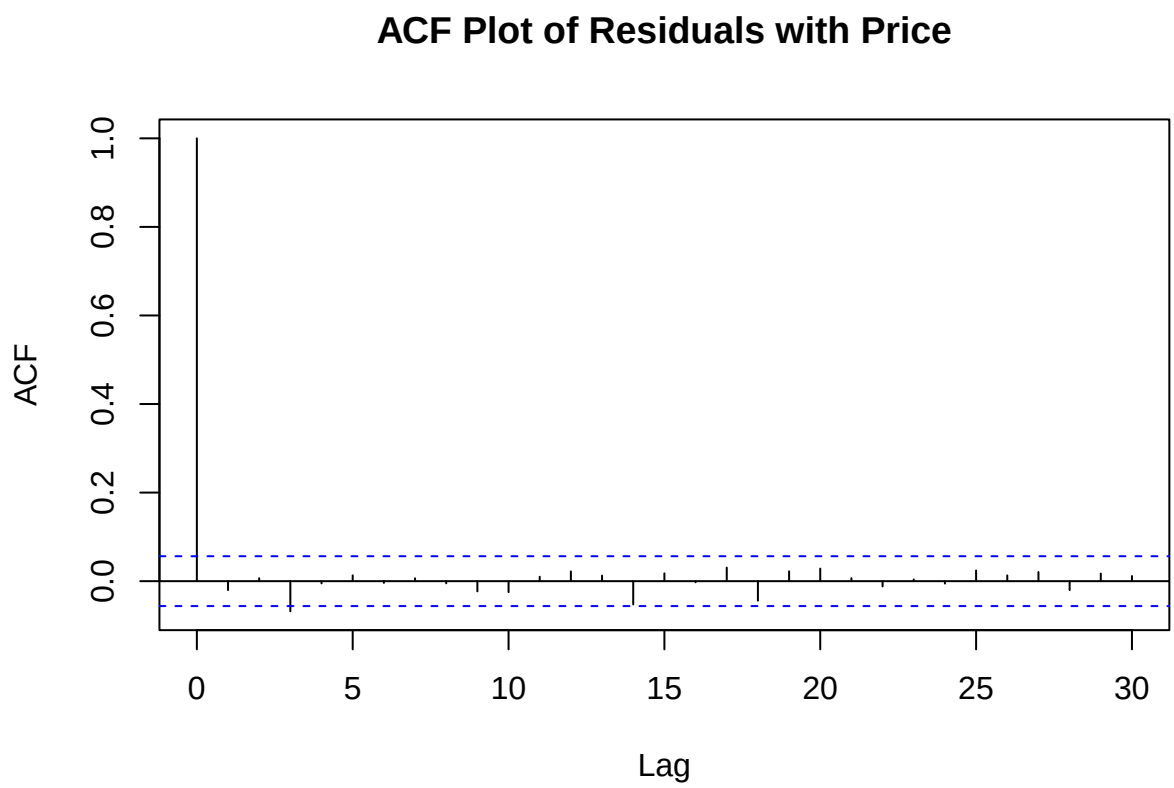
Additionally, the assumption of constant variance also does not look to hold true as the variance seems to be higher for higher values of carat. This means statistical inference test for hypothesis tests and confidence intervals can not be used until variance is stabilized.



Looking at the "Residual vs Fitted" plot, the vertical spread seems to be higher for higher fitted values indicating variance is not constant. Furthermore, the plot seems to indicate a non-linear relationship which indicates mean error is 0. Therefore, the first transformation we will try is on the response variable (Price). This will stabilize the variance of the response so we can assess assumption of mean error is 0 more accurately.

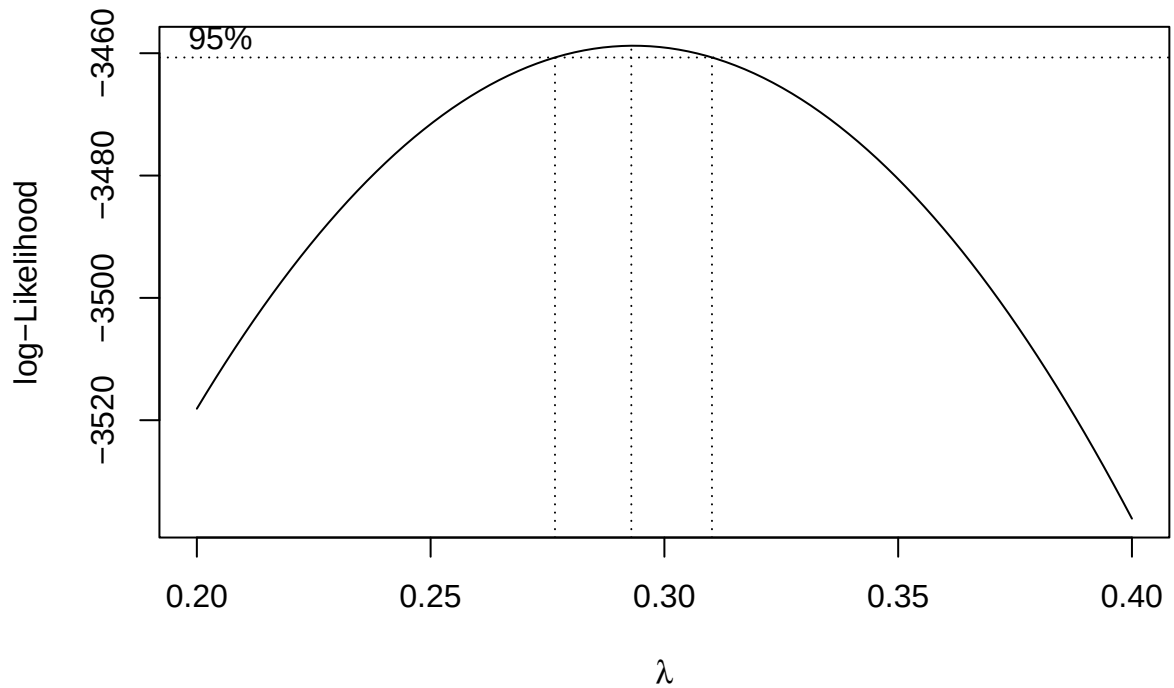
The "Scale-Location" plot also contributes additional evidence against the assumptions that variance is not constant as the plot shows curved fitted line and increasing variances moving left to right on the graph.

The "Q-Q Residuals" plot indicates that that errors are not normally distributed because the values do not fall on a roughly 45 degree line. However, since this assumption is the least important, it is ignored until the other assumptions are met.



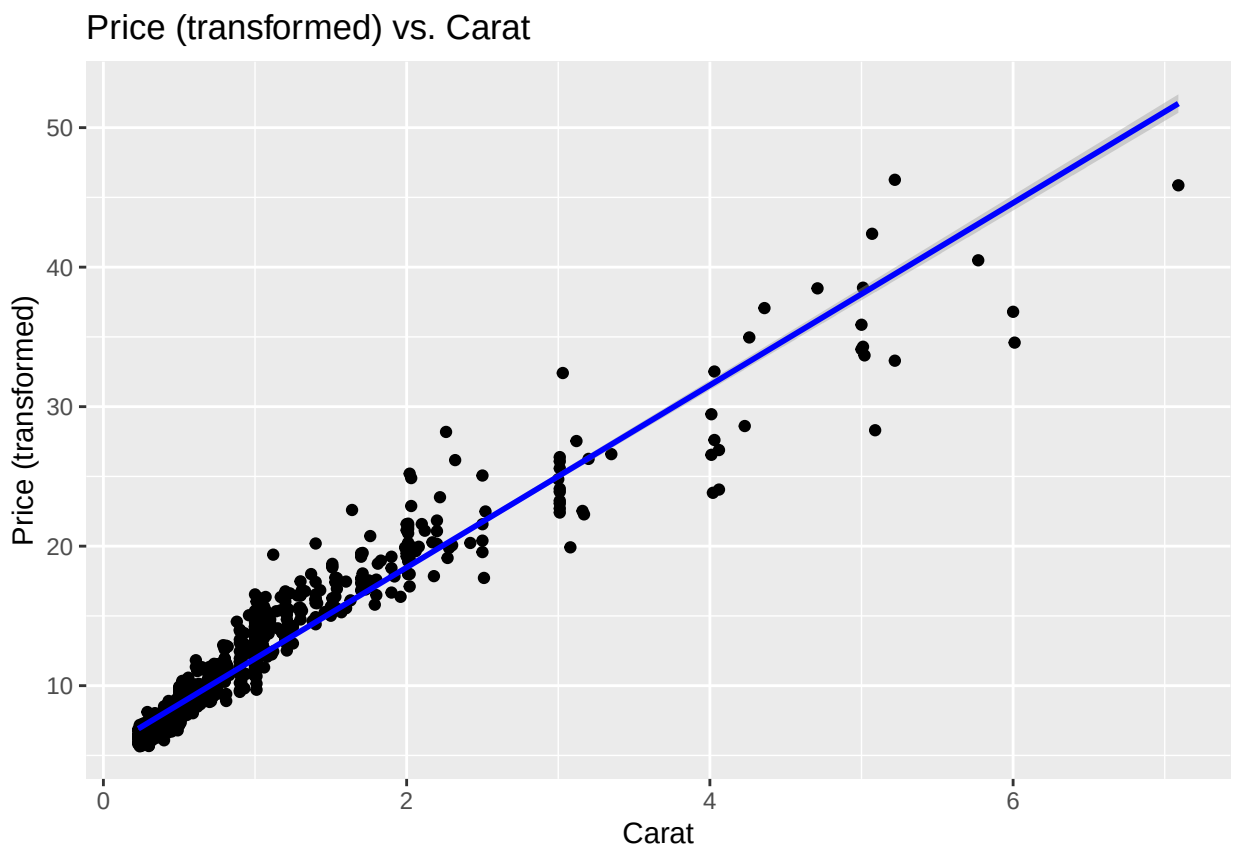
Given the "ACF Plot" the assumptions that errors are independent seems to hold true as lag is mostly under the critical value (expect for value 3).

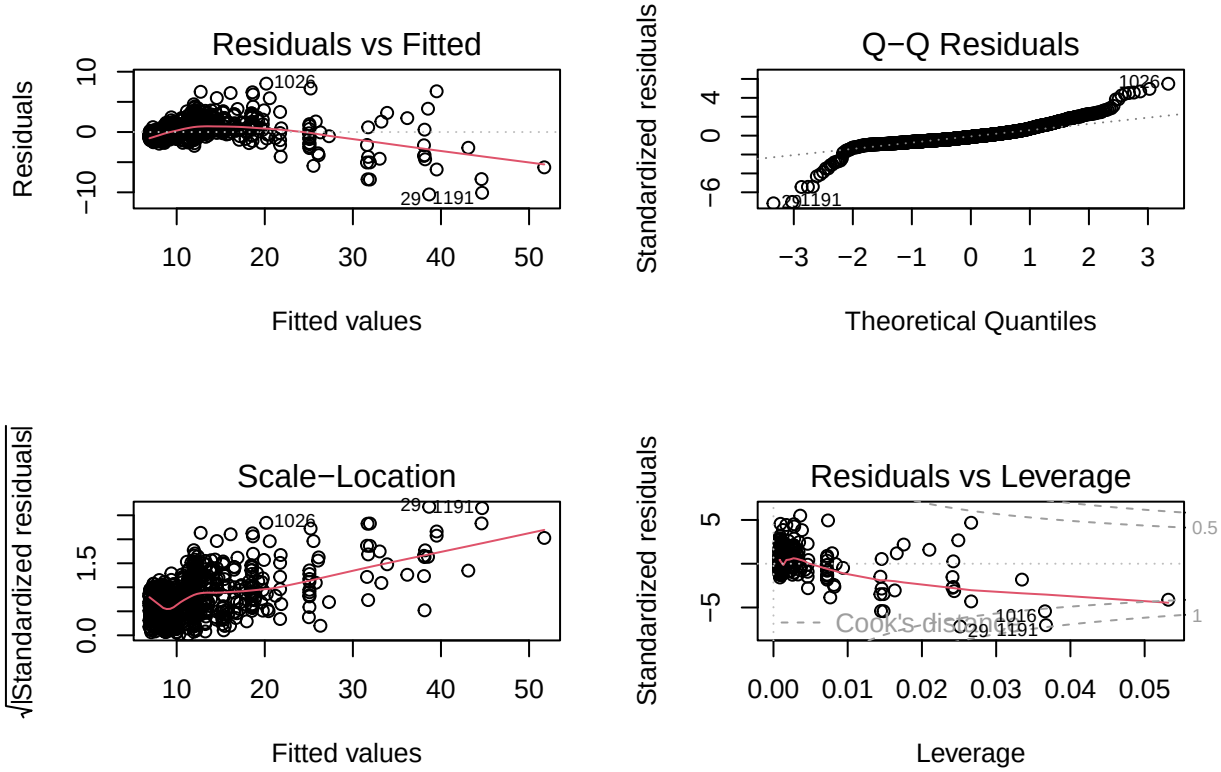
Transformation of variables



The box-cox plot maximizes log-likelihood function against lambda to determine which transformation will likely stabilize the variance of the response. Using a value inside the outer dashed lines (95% CI) is the most likely transformation of the data to stabilize variance. I will choose $\lambda = .3$ since it is inside the 95% CI and closest to the predicted optimal value for lambda. Log transformation would have been preferred because log transformation preserve the meaning of the coefficients. However, since $\lambda = 0$ is not in the 95% CI, this is not a viable transformation.

```
## `geom_smooth()` using formula = 'y ~ x'
```


Price (transformed) vs. Carat
Price (transformed)
50
40
30
20
10
0
2
4
6
Carat

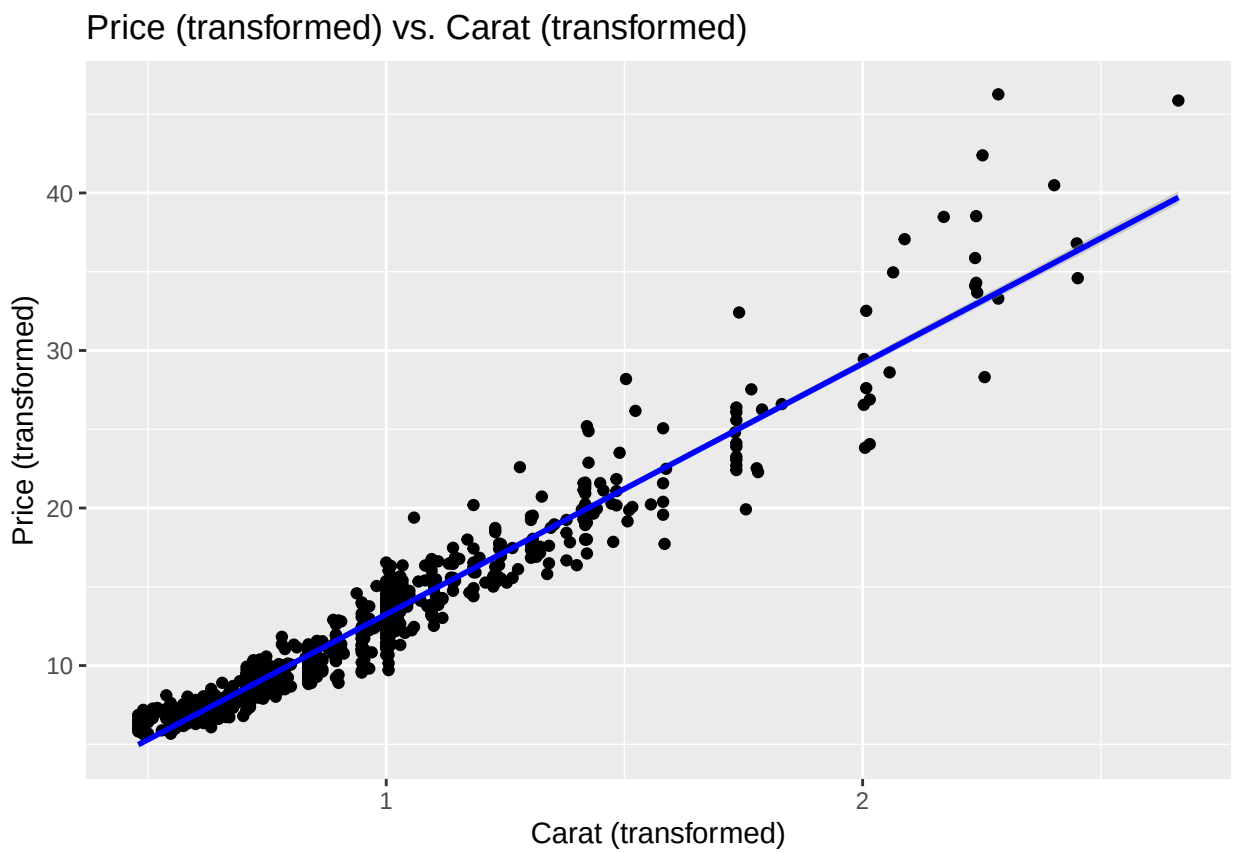


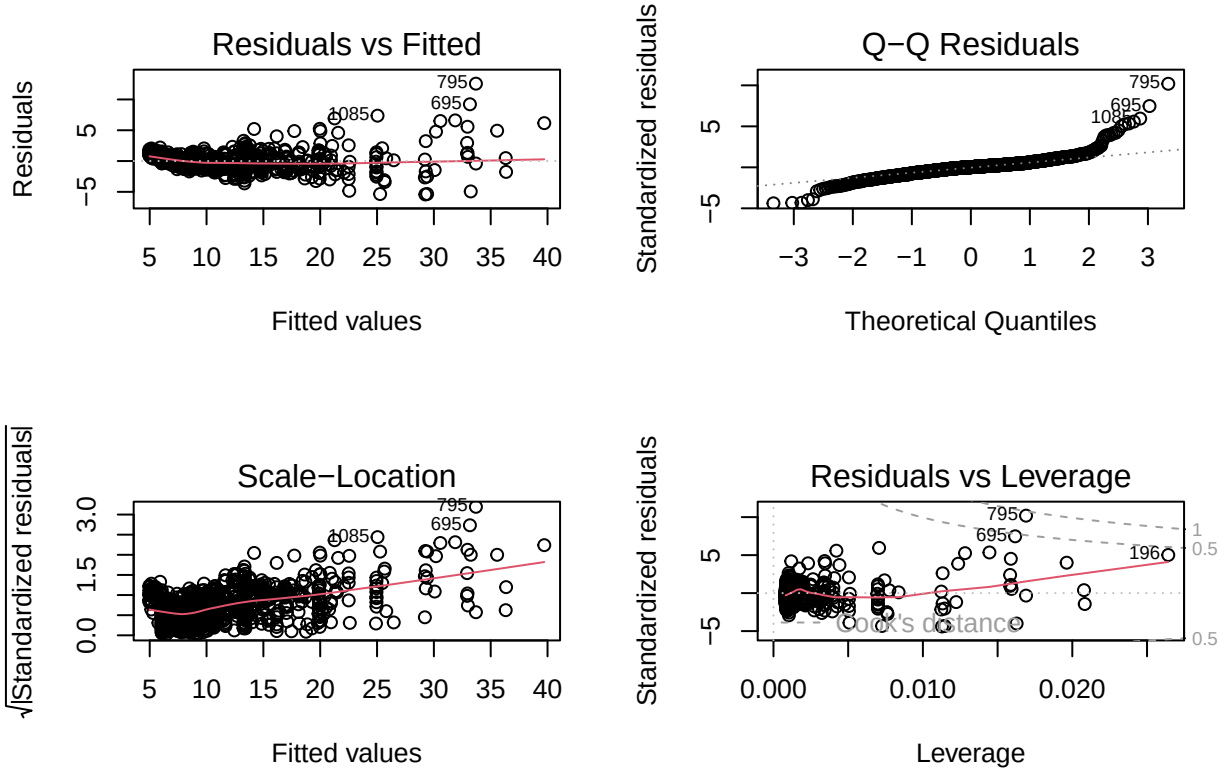
Looking at the "Residual vs Fitted" and "Scale-Location" plot, the variance of data around the fitted line is relatively constant moving left to right on the plot. This indicates the transformation on the response succeeded in stabilizing the variance so the constant variance of errors is met.

However, the "Residual vs Fitted" plot still seems to indicate a non linear relationship because the red line is not straight so a transformation on the predictor is the next step.

The un-transformed predictor plot above looks roughly like a square-root plot so we will attempt that transformation and assess residuals.

```
## `geom_smooth()` using formula = 'y ~ x'
```

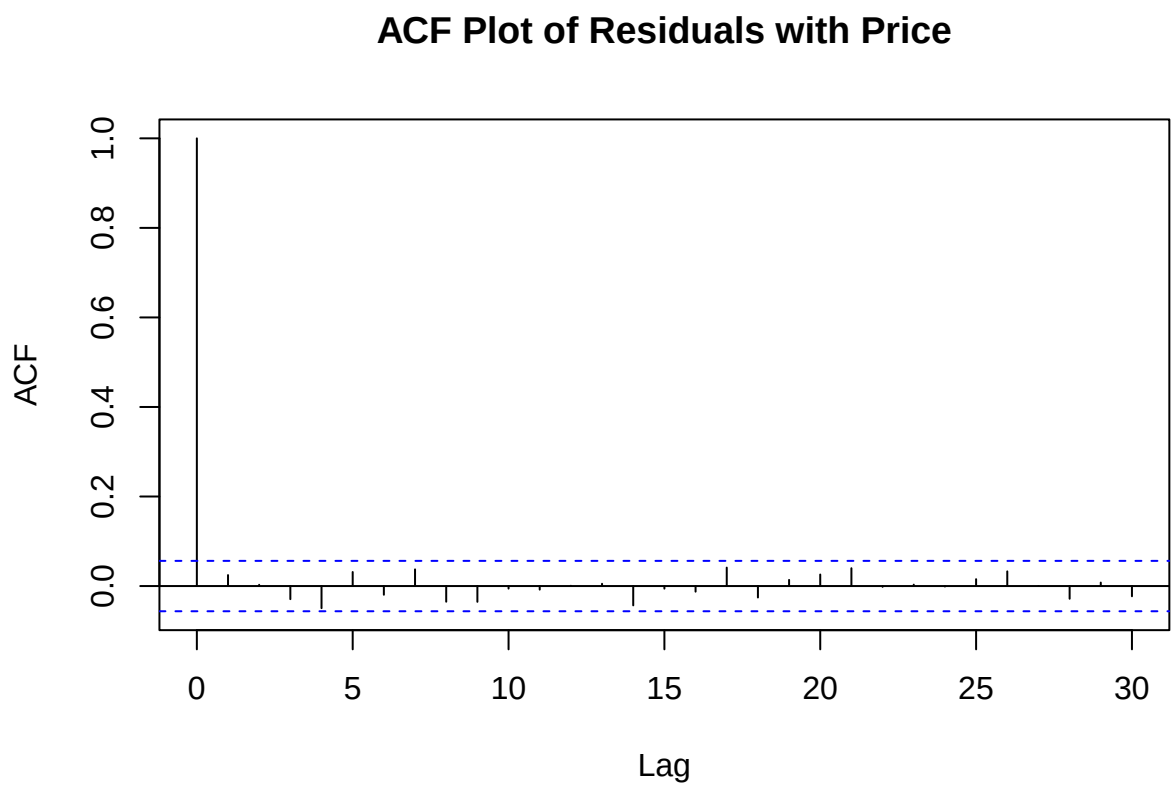

Price (transformed) vs. Carat (transformed)
Price (transformed)
40
30
20
10
1
2
Carat (transformed)



The square root transformation on the predictor seems to have improved the "Residual vs Fit" line so that it is straight and horizontal indicating that assumption mean error is 0 is met.

The assumption of constant variance seems to be plausibly met as the variance of the data in the "Residual vs Fit" graph is constant over the values of the response variable. Also the "Scale-Location" line has a slight slope, but this still supports the conclusion assumption of constant variance is met.

The "Q-Q Residuals" plot indicates that that errors are closer to a normal distribution after the transformation by comparing the plot before vs after the transformation, the values are closer to the 45 degree line with some variation in each end. However, since this assumption is the least important, we can make the assumption that it is met.



Given the "ACF Plot", the assumptions that errors are independent is met as all lag values are under the critical value.

Now that the four regression assumptions are met, we can fit a Simple Linear Regression model to our transformed data.

Simple Linear Regression

```
##
## Call:
## lm(formula = ystar ~ xstar, data = Data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.4231 -0.5646  0.0264  0.4750 12.5619
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2.65220    0.09844  -26.94   <2e-16 ***
## xstar       15.91060    0.10915  145.77   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.243 on 1212 degrees of freedom
## Multiple R-squared:  0.946,  Adjusted R-squared:  0.946
## F-statistic: 2.125e+04 on 1 and 1212 DF,  p-value: < 2.2e-16
```

Simple Linear Regression equation (where :

$\hat{Price}^{.3} = -2.65220 + 15.91060 * Carat^{1/2}$

Unfortunately the value of the parameters can not be interpreted because the transformation were not logarithmic. However, contextually the positive slope indicates that as the carat increase, the price increase. This relationship is non linear as seen by the transformations applied in the previous section. From the transformation applied to predictor it seems that as carat increase, price increases at a polynomial rate (i.e. the square root transformation to linearize the data).

After viewing the results of the hypothesis test, we can conclude that the results for (transformed) slope and intercept are statistically significant because they have p value much less than .05 ($p < 2 * 10^{-16}$).

The confidence intervals for slope are as follows:

```
##    2.5 %   97.5 %
## 15.69645 16.12474
```

The 95% confidence interval for slope is (15.69645, 16.12474) confirming the hypothesis test that the (transformed) relationship between Price and Carat is linear because the Confidence Interval does not include 0.